For the use of a Registered Medical Practitioner or a Hospital or a Laboratory only OR for Specialist Use only

Palonco 0.25mg / 5mL Solution for Injection
Palonosetron Hydrochloride

COMPOSITION

Each 5 ml of solution contains 0.2808mg of Palonosetron hydrochloride equivalent to 0.25mg of Palonosetron.

DOSAGE FORM

Intravenous injection

PRODUCT DESCRIPTION

Clear colourless solution

PHARMACOLOGY

Pharmacodynamics

Pharmacotherapeutic group: Antiemetics and antinauseants, serotonin ($5HT_3$) antagonists.
ATC code: A04AA05
Palonosetron is a selective high-affinity receptor antagonist of the $5HT_3$ receptor.

Mechanism of Action

Palonosetron is a $5\text{-}HT_3$ receptor antagonist with a strong binding affinity for this receptor and little or no affinity for other receptors.

Cancer chemotherapy may be associated with a high incidence of nausea and vomiting, particularly when certain agents, such as cisplatin, are used. $5\text{-}HT_3$ receptors are located on the nerve terminals of the vagus in the periphery and centrally in the chemoreceptor trigger zone of the area postrema. It is thought that chemotherapeutic agents produce nausea and vomiting by releasing serotonin from the enterochromaffin cells of the small intestine and that the released serotonin then activates $5\text{-}HT_3$ receptors located on vagal afferents to initiate the vomiting reflex.

Postoperative nausea and vomiting is influenced by multiple patient, surgical and anesthesia related factors and is triggered by release of 5-HT in a cascade of neuronal events involving both the central nervous system and the gastrointestinal tract. The $5\text{-}HT_3$ receptor has been demonstrated to selectively participate in the emetic response

The effect of palonosetron on blood pressure, heart rate, and ECG parameters including QTc were comparable to ondansetron and dolasetron in CINV. In PONV, the effect of palonosetron on the QTc interval was no different from placebo. Palonosetron possesses the ability to block ion channels involved in ventricular de- and re-polarisation and to prolong action potential duration.

No significant effect on any ECG interval including QTc duration (cardiac repolarization) at doses up to 2.25 mg.

Pharmacokinetics

Absorption

Following intravenous administration, an initial decline in plasma concentrations is followed by slow elimination from the body. Mean maximum plasma concentration (C_{max}) and area under the concentration-time curve ($AUC_{0-\infty}$) are generally dose-proportional over the dose range of 0.3–90 μg/kg.

After intravenous dosing of palonosetron in patients undergoing surgery (abdominal surgery or vaginal hysterectomy), the pharmacokinetics characteristics of palonosetron were similar to those observed in cancer patients.

Distribution

Palonosetron has a volume of distribution of approximately 8.3 +/- 2.5 L/kg. Approximately 62 % of palonosetron is bound to plasma proteins.

Metabolism

Palonosetron is eliminated by multiple routes with approximately 50% metabolised to form two primary metabolites: N-oxide-palonosetron and 6-S-hydroxy-palonosetron. These metabolites each have less than 1 % of the $5HT_3$ receptor antagonist activity of palonosetron. It was shown that CYP2D6 and to a lesser extent, CYP3A4 and CYP1A2 are involved in the metabolism of palonosetron. However, clinical pharmacokinetic parameters are not significantly different between poor and extensive metabolisers of CYP2D6 substrates.

Elimination

After a single intravenous dose of 10 micrograms/kg [^{14}C]-palonosetron, approximately 80 % of the dose was recovered within 144 hours in the urine with palonosetron representing approximately 40 % of the administered dose. Mean terminal elimination half-life is approximately 40 hours.

Special populations

Paediatric Patients

No apparent differences in volume of distribution when expressed in L/kg.

INDICATIONS

Palonosetron hydrochloride solution for injection is indicated for:

Chemotherapy-Induced Nausea and Vomiting

Adults and Paediatric Patients 1 month of Age and Older

- The prevention of acute nausea and vomiting associated with highly emetogenic cancer chemotherapy,
- The prevention of nausea and vomiting associated with moderately emetogenic cancer chemotherapy.

Postoperative Nausea and Vomiting

- Prevention of postoperative nausea and vomiting (PONV) for up to 24 hours following surgery. Efficacy beyond 24 hours has not been demonstrated.

As with other antiemetics, routine prophylaxis is not recommended in patients in whom there is little expectation that nausea and / or vomiting will occur postoperatively. In patients where nausea and vomiting must be avoided during the postoperative period, Palonosetron hydrochloride solution for injection is recommended even where the incidence of postoperative nausea and / or vomiting is low.

DOSAGE AND METHOD OF ADMINISTRATION

This medicinal product should be administered by a healthcare professional under appropriate medical supervision.

Posology

<u>Adults</u>

Chemotherapy-Induced Nausea and Vomiting

250 micrograms palonosetron administered as a single intravenous bolus approximately 30 minutes before the start of chemotherapy. Palonosetron hydrochloride solution for injection should be injected over 30 seconds.

The efficacy of palonosetron hydrochloride solution for injection in the prevention of nausea and vomiting induced by highly emetogenic chemotherapy may be enhanced by the addition of a corticosteroid administered prior to chemotherapy.

Postoperative Nausea and Vomiting

75 micrograms palonosetron administered as a single intravenous bolus over 10 seconds immediately before the induction of anaesthesia.

<u>Paediatric Population</u>

Chemotherapy-Induced Nausea and Vomiting

Children and Adolescents (aged 1 month to 17 years):

20 micrograms / kg (the maximum total dose should not exceed 1500 mcg) palonosetron administered as a single 15 minutes intravenous infusion beginning approximately 30 minutes before the start of chemotherapy.

The safety and efficacy of Palonosetron hydrochloride solution for injection in children aged less than 1 month have not been established. No data are available.

Postoperative Nausea and Vomiting
Safety and effectiveness in patients below the age of 18 years have not been established.

Elderly people
No dose adjustment is necessary for the elderly.

Hepatic impairment
No dose adjustment is necessary for patients with impaired hepatic function.

Renal impairment
No dose adjustment is necessary for patients with impaired renal function.
No data are available for patients with end stage renal disease undergoing haemodialysis.

Method of administration
For intravenous use.

CONTRAINDICATIONS
Hypersensitivity to the active substance or to any of the excipients

WARNINGS AND PRECAUTIONS
Hypersensitivity
Hypersensitivity reactions may occur in patients who have exhibited hypersensitivity to other 5-HT_3 receptor antagonists.

Serotonin Syndrome
Symptoms associated with serotonin syndrome may include the following combination of signs and symptoms: mental status change (e.g. agitation, hallucinations, delirium, and coma), autonomic instability (e.g. tachycardia, labile blood pressure, dizziness, diaphoresis, flushing, hyperthermia), neuromuscular symptoms (e.g.tremor, rigidity, myoclonus, hyperreflexia, incoordination), seizures, with or without gastrointestinal symptoms (e.g., nausea, vomiting, diarrhea). Patients should be monitored for the emergence of serotonin syndrome, especially with concomitant use of Palonosetron hydrochloride solution for injection and other serotonergic drugs (e.g selective serotonin reuptake inhibitors (SSRIs), serotonin and norepinephrine reuptake inhibitors (SNRIs), monoamine oxidase

inhibitors, mirtazapine, fentanyl, lithium, tramadol and intravenous methylene blue). If symptoms of serotonin syndrome occur, discontinue Palonosetron hydrochloride solution for injection and initiate supportive treatment. Patients should be informed of the increased risk of serotonin syndrome, especially if Palonosetron hydrochloride solution for injection is used concomitantly with other serotonergic drugs (see *Section Interactions with Other Medicaments)*.

INTERACTIONS WITH OTHER MEDICAMENTS

Palonosetron is eliminated from the body through both renal excretion and metabolic pathways with the latter mediated via multiple CYP enzymes. Palonosetron is not an inhibitor of CYP1A2, CYP2A6, CYP2B6, CYP2C9, CYP2D6, CYP2E1 and CYP3A4/5 nor does it induce the activity of CYP1A2, CYP2D6, or CYP3A4/5. Therefore, the potential for clinically significant drug interactions with palonosetron appears to be low.

<u>Serotonergic Drugs (e.g. SSRIs and SNRIs)</u>

Serotonin syndrome (including altered mental status, autonomic instability, and neuromuscular symptoms) has been described following the concomitant use of 5-HT_3 receptor antagonists and other serotonergic drugs, including selective serotonin reuptake inhibitors (SSRIs) and serotonin and noradrenaline reuptake inhibitors (SNRIs) (see *Section Warnings and Precautions)*.

<u>Corticosteroids</u>

Palonosetron has been administered safely with corticosteroids.

<u>Antiemetic / Antinauseants</u>

No significant interaction between palonosetron and oral aprepitant and oral metoclopramide.

<u>Chemotherapeutic agents</u>

Palonosetron did not inhibit the antitumour activity of the five chemotherapeutic agents tested (cisplatin, cyclophosphamide, cytarabine, doxorubicin and mitomycin C).

<u>Other medicinal products</u>

Palonosetron has been administered safely with analgesics, antispasmodics and anticholinergic medicinal products.

FERTILITY, PREGNANCY AND LACTATION

<u>Pregnancy</u>

Teratogenic Effects: Category B

No adequate and well-controlled studies in pregnant women. Therefore, palonosetron should be used during pregnancy only if clearly needed.

Labor and Delivery

Palonosetron has not been administered to patients undergoing labor and delivery, so its effect on the mother or child are unknown.

Breast feeding

It is not known whether palonosetron is excreted in human milk. Because many drugs are excreted in human milk and because of the potential for serious adverse reactions in nursing infants and the potential for tumorigenicity shown for palonosetron, a decision should be made whether to discontinue nursing or to discontinue the drug, taking into account the importance of the drug to the mother.

UNDESIRABLE EFFECTS

Chemotherapy-Induced Nausea and Vomiting

Adults

The most common adverse reactions were headache and constipation.

The following adverse reactions were reported for palonosetron.

System organ class	**Common ARs**	**Uncommon ARs**
Cardiovascular	Non-sustained Tachycardia, bradycardia, hypotension	Hypertension, myocardial ischaemia, extrasystoles, sinus tachycardia, sinus arrhythmia, supraventricular extrasystoles and QT prolongation
Dermatological		Allergic dermatitis, rash
Hearing and Vision		Motion sickness, tinnitus, eye irritation and amblyopia.
Gastrointestinal System	Constipation, Diarrhoea	Dyspepsia, abdominal pain, dry mouth, hiccups, flatulence
General	Weakness	Fatigue, fever, hot flush, flu-like syndrome
Liver		Transient, asymptomatic increases in AST and / or ALT and bilirubin. These changes occurred predominantly in patients receiving highly emetogenic chemotherapy.

Metabolism	Hyperkalaemia	Electrolyte fluctuations, hyperglycemia, metabolic acidosis, glycosuria, appetite decrease, anorexia
Musculoskeletal		Arthralgia
Nervous system	Headache, Dizziness	Somnolence, insomnia, paraesthesia, hypersomnia
Psychiatric	Anxiety	Euphoric mood
Urinary system		Urinary retention
Vascular disorders		Vein discolouration, vein distended

Paediatric population

The following adverse reactions were reported for palonosetron.

System organ class	**Common ARs**	**Uncommon ARs**
Nervous system		Headache, Dizziness, dyskinesia
General		Infusion site pain
Dermatological		Allergic dermatitis, skin disorder

Postoperative Nausea and Vomiting

Adults

The following adverse reactions were reported for palonosetron.

System organ class	**Common ARs**	**Uncommon ARs**
Cardiovascular	Electrocardiogram QTc prolongation, sinus bradycardia, tachycardia	Blood pressure decreased, hypotension, hypertension, arrhythmia, ventricular extrasystoles, generalized edema; ECG T wave amplitude decreased, platelet count decreased.
Dermatological	Pruritus	
Gastrointestinal System	Flatulence	Dry mouth, upper abdominal pain, salivary hypersecretion, diarrhea, intestinal hypomotility, anorexia
General		Chills
Liver	Increase in AST and / or ALT	Hepatic enzyme increased

Metabolic		Hypokalemia, anorexia
Nervous system		Dizziness
Respiratory		Hypoventilation, laryngospasm
Urinary system	Urinary retention	

Very rare cases: Hypersensitivity reactions and injection site reactions (burning, induration, discomfort and pain).

<u>Reporting of suspected adverse reactions</u>
Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the local reporting system.

OVERDOSE
There is no known antidote to Palonosetron hydrochloride solution for injection. Overdose should be managed with supportive care.

Doses of up to 6 mg have been used in adult. Due to the large volume of distribution, dialysis is unlikely to be an effective treatment for palonosetron overdose. The major signs of toxicity were convulsions, gasping, pallor, cyanosis and collapse.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES
No studies on the effects on the ability to drive and use machines have been performed.
Since palonosetron may induce dizziness, somnolence or fatigue, patients should be cautioned when driving or operating machines.

INSTRUCTIONS FOR USE
Single use only, any unused solution should be discarded.
Any unused product or waste material should be disposed of in accordance with local requirements.

INCOMPATIBILITY
This medicinal product must not be mixed with other medicinal products

STORAGE CONDITION
Store below 30°C. Protect from light.

SHELF LIFE
24 months
Upon opening of the vial, use immediately and discard any unused solution.

PACKAGING INFORMATION
Palonco 0.25 mg / 5 ml Solution for Injection is packed in 1 carton having 1 vial of 5ml USP type I flint (clear) glass vial

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